Year 12 Maths - Integration

Similar questions - Finding a curve from its gradient

Two new questions on this topic, a little more open-ended than the exam questions. They are not past exam questions. Answers are at the end.

1 A curve has gradient function $\frac{dy}{dx} = 3x^2 - 8x + 4$ and passes through $(1, 2)$.

(a) Find the equation of the curve. **[4]**

(b) Find the coordinates of the stationary points of the curve. **[4]**

(c) Hence explain why the equation $y = 0$ has only one real root. **[2]**

2 A curve passes through $(1, 2)$ and satisfies $\frac{dy}{dx} = 6\sqrt{x} - 2x$ for $x \geqslant 0$.

(a) Find the equation of the curve. **[4]**

(b) Find the equation of the tangent to the curve at the point where $x = 4$. **[3]**

(c) Show that this tangent meets the y-axis at the same point as the curve does. **[1]**

Question 1

(a) $y = x^3 - 4x^2 + 4x + c$. At $(1,2)$: $1 - 4 + 4 + c = 2$, so $c = 1$. $y = x^3 - 4x^2 + 4x + 1$

(b) $3x^2 - 8x + 4 = (3x - 2)(x - 2) = 0$: $x = \frac{2}{3}$ and $x = 2$. $\left(\frac{2}{3}, \frac{59}{27}\right)$ (maximum) and $(2,1)$ (minimum)

(c) Both stationary points have positive y-values, so the curve is above the x-axis between and around them: it can only cross the axis once, to the left of the maximum

Question 2

(a) $y = 4x^{3/2} - x^2 + c$. At $(1,2)$: $4 - 1 + c = 2$, so $c = -1$. $y = 4x^{3/2} - x^2 - 1$

(b) At $x = 4$: $y = 32 - 16 - 1 = 15$ and gradient $= 12 - 8 = 4$. $y - 15 = 4(x - 4)$, so $y = 4x - 1$

(c) The tangent meets the y-axis at $(0, -1)$. The curve at $x = 0$ gives $y = -1$, the same point